Risk in a Shared Public–Private Pool

Giuseppe Paleologo

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The Questions

Our main fund has both privates and publics in a shared pool (not side-pocketed). How should we conceptually think about risk given this?

- Should we manage risk in the public book to overall NAV, or just public NAV, or some mixture? How does this vary with different % private, and different timelines to liquidity?
- What are the risk considerations we should take into account when deciding on a marginal private investment? How does this vary with different % private?
  - How should we think about the risk and liquidity considerations between privates that are close to IPO and likely to succeed vs. more call-option-y earlier stage privates?
  - What does that imply for the IRR threshold for a private vs. a public (taking into account cost of liquidity, cost of risk, and leverage constraints)?

Simplified constraints/setting:

- Capital is locked up for a long time, and if necessary we can put things in "liquidating capital accounts" (i.e. redeem in kind rather than in cash).
  - So there can't be an inability to pay / run on the bank issue.
  - That said, broadly the expectation is that folks will be able to redeem in cash in normal circumstances (where there is only a small percentage redeeming at a given time), so it's costly for us if we end up with a very high % illiquid.
    - High illiquidity (say, >50%) introduces further risks such as valuation and potentially broader investor frustration
- People see and judge an overall performance number on overall NAV, so most of the cost of drawdowns from an IR etc. perspective is on the overall NAV. (Below some limit, such as the <50% above)
- We've generally guided to 20–30% private, as target allocations, but not hard constraints.
- Representatively, consider:
  - **Private A** (mega cap private): Very likely to go public in the next 12 months, like a large cap stock. (Ex.: Anthropic)
  - **Private B** (late stage private): Likely to go public in the next 12–36 months, more risky than A but successful in most worlds. (Ex: Fluidstack)
  - **Private C** (early stage private): Isn't successful in 70% of worlds, in 30% of worlds is wildly successful (a bit like a call option). Will take a few years to prove out. (Ex.: Source Foundry)

- For thinking about how to think about the risk of the public sleeve of the book, consider
  - **Setting 1:** Fund is 15% private (10% Private B, 5% Private C)
  - **Setting 2:** Fund is 60% private (45% Private A, 10% Private B, 5% Private C).
    - In this setting, how much more volatility (relative to public NAV) can we take on the currently-public sleeve of the book? How close to managing to a denominator of overall NAV vs. just the public sleeve should we be?
- Broadly think of bottom line utility of the fund as just log($) in the long run
- We can fundraise (bringing in additional liquid $ reducing % private)
  - Though fundraising is harder when you've lost a lot of money and/or privates are extremely high.

Measure and size risk on total economic NAV, and treat illiquidity as two separate things: a smaller effective denominator for the public book's alpha, and a cash constraint along adverse paths. At 15% private, the public book should be managed to total NAV (about 1.13× the vol an all-public fund would run). At 60% private, the public book should first hedge the privates' theme exposure; its alpha can then run about 1.3–2.2× the all-public level, and a marginal private needs roughly 5–30 more points of expected return than at 15%.

All numbers below come from stipulated profiles for A, B and C and from stated parameters. They show the mechanism and the orders of magnitude; they are not forecasts about the named companies.

The other answer improves on mine in an important way: **it turns the liquidity problem into an explicit, state-dependent sizing rule.** I would adopt that approach.

But I would not adopt all its conclusions. Its liquidity-penalty derivation is sound; conclusions such as "manage fully to total NAV at 15%," "B is the most dangerous per dollar," and "selling A at a 5–10% discount is cheap" require assumptions that have not been established.

The central synthesis is:

> **Total NAV is the wealth whose growth you are optimizing. An effective capital base determines the appropriate size of independent public alpha. Economic exposures determine the public hedges. Usable liquidity constrains both.**

Those are four related concepts, not four competing definitions of NAV.

1. What the two answers share, where they differ, and what I would change

Issue	Common ground	Difference and revision
Whose wealth matters?	Both place the economic objective on total investor wealth, including privates.	I would retain this unequivocally. Private wealth does not stop counting because it cannot currently be sold.
Independent public alpha	My first answer noted that independent risks separate in the frictionless log-utility benchmark. The alternative makes this central.	**I would make it central too.** My subsequent volatility-budget example gave too much prominence to a fixed fund-volatility ceiling, which is an additional policy choice—not a consequence of log utility.

Issue	Common ground	Difference and revision
Liquidity and public sizing	Both recognize that liquidity can constrain public risk-taking.	The alternative supplies a useful explicit model for an effective denominator. I would incorporate it, while distinguishing an illustrative calibration from a recommended limit.
Hedging privates	Both recommend evaluating public positions jointly with private exposures.	I would separate **public alpha sizing** from **public hedging** much more explicitly. A hedge can increase public-sleeve volatility while reducing whole-fund risk.
Marginal private investments	Both identify economic risk, liquidity, and foregone public opportunities as relevant costs.	I would retain my full-distribution log-utility test. Covariance is a local diffusion risk charge, not an exact general formula for binary private investments.
Numerical conclusions	Both use illustrative assumptions.	I would not infer universal stage rankings, return hurdles, or secondary-sale decisions from those assumptions. The alternative's multiplier table checks out; its investment conclusions are much more calibration-dependent.

Three further corrections matter.

First, **15% is not a theoretical boundary**. Being well below a liquidity threshold today does not mean that liquidity has zero prospective cost.

Second, **a blended denominator need not lie between public NAV and total NAV**. Severe liquidity constraints can justify taking less public alpha risk than a stand-alone public fund would take on the same liquid capital.

Third, the distinction between a **whole position's attractiveness** and the attractiveness of **adding another dollar to that position** is important. The alternative partially recognizes this, but its early-stage IRR discussion mixes quantities that should be kept separate.

Below is the framework I would now use.

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2. Define the objective and the relevant balance sheets

Let

$$N = L + P,$$

where:

- N is total **economic** NAV;
- L is the net value of the liquid portfolio, including cash;
- P is the economic value of the private holdings.

Define the private fraction as

$$\pi = \frac{P}{N}.$$

Also maintain **reported NAV**, denoted $\widetilde{N}$, and reported private value $\widetilde{P}$. Their ratio,

$$\widetilde{\pi} = \frac{\widetilde{P}}{\widetilde{N}},$$

matters for investor communications and reported allocation limits.

Economic and reported values should not be conflated. Illiquid-asset return smoothing can make reported volatility lower than underlying economic volatility. That is a documented measurement problem, not genuine diversification. (nber.org)

Finally, liquid NAV is not necessarily the same as **usable liquidity**. Some assets may be pledged, some cash may support margin, and some nominally liquid positions may be expensive to unwind.

The objective

Taking your preference literally, the investment objective is

$$\max \mathbb{E}[\log W_T],$$

where W_T is the wealth of a continuing investor.

With no subscriptions or redemptions, we can work with fund NAV:

$$\max \mathbb{E}[\log N_T].$$

With capital flows, use investor wealth or flow-adjusted NAV per share. **Fundraising is not investment performance**, although it can change the fund's financing capacity.

Your other considerations enter in two ways.

Hard constraints determine what the fund can finance: margin, contractual obligations, operating cash, and whatever normal cash-redemption capacity you choose to preserve.

Economic or commercial costs capture consequences such as losing investment flexibility or triggering investor frustration that affects future economics.

A reduced-form objective might therefore be

$$\max \mathbb{E}\left[\log W_T - \int_0^T c(\widetilde{\pi}_t)\, dt\right],$$

subject to financing constraints.

Here c is a cost measured in log-wealth-equivalent units per year.

There is an important discipline: **do not add a frustration penalty merely because illiquidity feels uncomfortable.** It should represent consequences you actually care about, and should not duplicate costs already included in cash flows or investment returns. If frustration has no effect on investor wealth and is not a separate objective, a pure log-wealth model should not charge for it.

Takeaway: distinguish investment wealth, reported performance, usable liquidity, and any additional commercial objectives before choosing risk limits.

3. Start with the frictionless log-utility benchmark

This establishes what illiquidity changes.

Let u be public risky exposures expressed as fractions of total NAV, and y the private weights. Let μ_U, μ_P be expected excess returns, and partition the instantaneous covariance matrix as

$$\Sigma = \begin{pmatrix} \Sigma_{UU} & \Sigma_{UP} \\ \Sigma_{PU} & \Sigma_{PP} \end{pmatrix}.$$

In a diffusion model, instantaneous expected log growth is

$$g = r_f + u^\top \mu_U + y^\top \mu_P - \frac{1}{2}\left(u^\top \Sigma_{UU} u + 2u^\top \Sigma_{UP} y + y^\top \Sigma_{PP} y\right).$$

Holding the private positions fixed at the current instant, differentiate with respect to public exposures:

$$\frac{\partial g}{\partial u} = \mu_U - \Sigma_{UU} u - \Sigma_{UP} y.$$

Thus the unconstrained, locally growth-optimal public portfolio is

$$\boxed{u^* = \Sigma_{UU}^{-1} \mu_U - \Sigma_{UU}^{-1} \Sigma_{UP} y.}$$

This separates two jobs.

The first term represents public investment opportunities. The second adjusts public positions for exposures already held privately.

Independent alpha does not automatically lose risk budget when privates increase

Suppose a public alpha strategy has innovations independent of the private holdings and relevant state variables. Its covariance with private returns is zero.

In this benchmark, its optimal exposure does not fall merely because private variance rises:

$$u_\alpha^* = \frac{\mu_\alpha}{\sigma_\alpha^2}.$$

Its **dollar** exposure is therefore

$$q_\alpha^* = N \frac{\mu_\alpha}{\sigma_\alpha^2}.$$

This is the precise sense in which it is sized against total NAV.

Independent private risk still changes the distribution of total fund returns. But it does not “use up” a preassigned variance budget unless you separately impose such a budget.

That distinction is the main change I would make to my earlier presentation.

Why this is only a benchmark

The calculation maximizes **instantaneous** expected log growth. It does not solve the full problem when private holdings cannot be rebalanced, liquidity dates are uncertain, and future financing constraints depend on today's returns.

Dynamic illiquid-asset models can reduce liquid risky allocations even with log utility and uncorrelated asset returns. For example, Ang, Papanikolaou, and Westerfield obtain this result when liquid wealth must finance consumption. Your fund's cash needs differ, but the mechanism—liquid wealth having special value—is relevant. (kellogg.northwestern.edu)

Takeaway: use total NAV for independent alpha in the frictionless benchmark. Departures from that rule should come from identified constraints, future-state effects, or additional costs—not from an arbitrary decision to exclude private wealth.

4. Derive an effective capital base for public alpha

This is the most useful addition from the alternative answer.

Let

$$J(L, P, t)$$

be the optimized value of the fund's future opportunities, given current liquid and private wealth.

Consider an independent public alpha strategy with dollar exposure q, expected excess return μ, and volatility σ per unit of exposure:

$$dL = q\mu\, dt + q\sigma\, dB_t.$$

Its contribution to the expected change in J is

$$J_L q\mu\, dt + \frac{1}{2} J_{LL} q^2 \sigma^2\, dt.$$

The first-order condition is

$$J_L \mu + J_{LL} q \sigma^2 = 0.$$

Therefore,

$$\boxed{q^* = D_{\text{eff}} \frac{\mu}{\sigma^2}, \qquad D_{\text{eff}} = -\frac{J_L}{J_{LL}}.}$$

The quantity D_{eff} is **risk tolerance with respect to liquid wealth**. It is the appropriate effective capital base for this independent alpha strategy.

It is not a valuation haircut on the private holdings.

If

$$J(L, P) = \log(L + P),$$

then

$$J_L = \frac{1}{N}, \qquad J_{LL} = -\frac{1}{N^2},$$

and hence

$$D_{\text{eff}} = N.$$

The total-NAV benchmark follows immediately.

Dynamic optimization with separate liquid and illiquid wealth is the broader framework in which this calculation belongs; random liquidation times are explicitly modeled in that literature. (arxiv.org)

A tractable approximation

Adopt the alternative answer's simplified continuation value:

$$J(L,P) = \log N - \Phi(\pi).$$

For this calculation, assume economic and reported values coincide, and freeze other state variables.

Since

$$\pi = \frac{P}{L+P}, \qquad \frac{\partial \pi}{\partial L} = -\frac{\pi}{N},$$

we obtain

$$J_L = \frac{1 + \pi\Phi'(\pi)}{N},$$

and

$$J_{LL} = -\frac{1 + 2\pi\Phi'(\pi) + \pi^2\Phi''(\pi)}{N^2}.$$

Therefore,

$$\boxed{D_{\text{eff}} = k(\pi)N, \qquad k(\pi) = \frac{1 + \pi\Phi'}{1 + 2\pi\Phi' + \pi^2\Phi''}.}$$

The alternative's formula is correct.

A liquid gain has extra value because it reduces the private fraction. A liquid loss has extra cost because it increases the private fraction. With a convex penalty, the additional downside sensitivity reduces alpha sizing.

You can express the result as

$$D_{\text{eff}} = L + \theta P,$$

where

$$\theta = \frac{k - (1-\pi)}{\pi}.$$

But θ should be understood as a **sizing coefficient**, not "the percentage of private value we believe is real."

5. Apply that model to your two settings

Use the illustrative penalty

$$c(\pi) = f_0 \left(\frac{(\pi - 0.30)_+}{0.30} \right)^2, \qquad \Phi(\pi) = \tau c(\pi).$$

Here:

- f_0 is the annual cost of remaining at 60% private;
- τ is an approximation to the period over which this cost persists.

The 30% threshold, quadratic shape, and f_0 are **policy assumptions**, not estimates supplied by the investment profiles.

Take

$$f_0 = 0.02.$$

This treats one year at 60% private as costing approximately 2% of NAV in wealth-equivalent terms.

Setting 1: 15% private

At the current state, the assumed penalty is flat:

$$\Phi' = \Phi'' = 0.$$

Therefore,

$$k = 1, \qquad D_{\text{eff}} = N.$$

Let σ_0 be the volatility of the same optimally sized independent alpha strategy in an all-liquid version of the fund.

Its volatility measured relative to public NAV becomes

$$\sigma_{\alpha,L} = \frac{N}{L}\sigma_0 = \frac{\sigma_0}{0.85} = 1.1765\,\sigma_0.$$

Thus, if the all-liquid alpha allocation would have 10% volatility, it would have approximately **11.8% volatility on the 85% liquid sleeve**, while generating the same dollar risk relative to total NAV.

That is a benchmark—not an unconditional recommendation.

For example, if public assets fall 40% while private marks remain unchanged,

$$\pi' = \frac{15}{85(0.60) + 15} = 22.7\%.$$

The private fraction does not remain near 15%. A larger public loss, redemption, or private appreciation could push it further.

Setting 2: 60% private

At $\pi = 0.60$,

$$\Phi' = \frac{2\tau f_0(0.60 - 0.30)}{0.30^2}.$$

With $f_0 = 0.02$ and $\tau = 1.5$,

$$\Phi' = 0.20, \qquad \Phi'' = \frac{2(1.5)(0.02)}{0.09} = 0.6667.$$

Consequently,

$$k = \frac{1 + 0.60(0.20)}{1 + 2(0.60)(0.20) + 0.60^2(0.6667)} = \frac{1.12}{1.48} = 0.7568.$$

For NAV of 100,

$$D_{\text{eff}} = 75.68.$$

Since liquid NAV is 40,

$$D_{\text{eff}} = 40 + 0.595(60).$$

Under these assumptions, the effective capital base is therefore approximately **public NAV plus 60% of private NAV**, rather than exactly half.

The alpha-volatility multiplier is

$$\frac{D_{\text{eff}}}{L} = \frac{75.68}{40} = 1.892.$$

Sensitivity to private share and duration

The following table uses $f_0 = 2\%$. The last column assumes the same alpha strategy would run at 10% volatility in an all-liquid fund.

Private share	Assumed duration τ	Effective capital / total NAV	Alpha volatility measured on liquid NAV
15%	Local penalty inactive	100.0%	**11.8%**
40%	3 years	79.8%	**13.3%**
50%	3 years	70.8%	**14.2%**
60%	1.5 years	75.7%	**18.9%**
60%	3 years	63.3%	**15.8%**
60%	5 years	53.8%	**13.5%**

For your two settings, this calibration permits approximately **61% more alpha volatility relative to liquid NAV** in Setting 2 than Setting 1 if the costly period lasts 1.5 years:

$$\frac{18.9\%}{11.8\%} \approx 1.61.$$

That increase falls to about 34% with a three-year duration.

These are **alpha-only results before financing constraints**, not total-public-book volatility limits.

What should replace the crude duration assumption?

Ideally, calculate

$$\Phi = \mathbb{E}\left[\int_0^T c(\pi_t)\,dt\right],$$

rather than multiplying today's cost by one assumed duration.

The paths should jointly model private values, public returns, liquidity dates, cash needs, and fundraising. Uncertainty about the duration of illiquidity is itself economically important in dynamic portfolio models. (nber.org)

For A, use the time until an economically meaningful amount is **actually tradeable or monetizable**, not just the IPO date. The SEC notes that lockups and other resale restrictions can prevent existing investors from selling immediately after listing, and that terms vary. (investor.gov)

Takeaway: the blended-denominator idea is useful, but the blend is an output. "Half the private NAV" is a possible calibration, not a principle.

6. Choose public hedges separately—and stress their cash needs

Suppose private returns have a theme component

$$r_i = \beta_i F + \varepsilon_i,$$

where F is the theme return and ε_i is company-specific return.

The private theme exposure, measured relative to total NAV, is

$$b_P = \sum_i y_i \beta_i.$$

If the desired total-fund theme exposure is b^*, the public portfolio should supply

$$\boxed{b_U = b^* - b_P.}$$

This is exposure accounting. Liquidity and investment views determine the desirable b^* and whether the hedge is feasible.

A numerical illustration

Assume, solely for this example,

$$\beta_A = 1, \qquad \beta_B = 1.5, \qquad \beta_C = 0,$$

and choose a total-fund target

$$b^* = 0.20.$$

Then:

	Setting 1	Setting 2
Private theme exposure	$0.10(1.5) = 0.15$	$0.45(1) + 0.10(1.5) = 0.60$
Required public theme exposure	$+0.05$	-0.40

In Setting 2, the public portfolio would be short theme exposure equal to 40% of total NAV—100% of liquid NAV.

That could make the public sleeve look volatile on its own while reducing the fund's economic risk.

The conclusion is conditional: **net short is appropriate when existing private exposure exceeds the desired whole-fund exposure**, not merely because private NAV is high.

A reconciled stress calculation

Normalize Setting 2 to NAV 100: liquid assets 40, private assets 60.

Assume:

- the theme falls 35%;
- private company-specific returns are zero in this scenario, producing a private loss of $0.60 \times 35\% = 21$;
- independent public alpha loses 1.5 times its annual dollar volatility;
- investors then withdraw 5 units in cash;
- the hedge, when present, is short 40 units of theme exposure and gains $40 \times 35\% = 14$.

Private assets end at 39.

Alpha volatility / initial liquid NAV	Theme hedge?	Economic investment loss	Liquid NAV after cash redemption	Private fraction afterward
10%	No	27%	29	57.4%
25%	No	36%	20	66.1%
10%	Yes	13%	43	47.6%
25%	Yes	22%	34	53.4%

For example, the final row is

$$\text{alpha loss} = 1.5(25\%)(40) = 15,$$

$$\text{investment loss} = 21 + 15 - 14 = 22,$$

$$L' = 40 - 15 + 14 - 5 = 34,$$

$$\pi' = \frac{39}{39 + 34} = 53.4\%.$$

The cash withdrawal is not counted as investment performance.

This illustrates the alternative answer's valid point: **hedging can matter more than a modest change in alpha sizing.**

But reverse the market scenario. A 35% theme rally costs the short hedge 14 units of liquid capital, even as the privates appreciate.

Therefore, do not assume that hedge losses are harmless because IPOs and fundraising will be available in up-markets. That is a scenario assumption, not a financing commitment.

Also, do not assume a particular direct or proxy hedge is permitted under the investment and lockup documents. Review the actual terms.

Takeaway: economic hedging and cash survival must both work. A hedge can improve one and worsen the other.

7. Evaluate a marginal private investment against what it actually displaces

The correct question is not "Is this a good private IRR?"

It is:

> **Does buying this amount of the private investment improve expected investor log wealth after changing the public portfolio and accounting for cash constraints?**

The exact terminal-payoff test

Normalize initial NAV to one.

Let G be baseline terminal fund wealth. Suppose a fraction δ is moved from an alternative with terminal gross payoff X_a into private investment j, with payoff X_j.

Then

$$G_{\text{new}} = G + \delta(X_j - X_a).$$

The exact change in expected log wealth is

$$\boxed{\Delta U = \mathbb{E}\left[\log\left(1 + \frac{\delta(X_j - X_a)}{G}\right)\right].}$$

For an infinitesimal investment,

$$\boxed{\left.\frac{\partial \Delta U}{\partial \delta}\right|_{\delta=0} = \mathbb{E}\left[\frac{X_j - X_a}{G}\right].}$$

Payoffs receive greater weight when the rest of the fund is poor.

For the actual fund, allow public positions to change and include path-dependent liquidity costs. The formula above isolates the terminal-payoff logic.

The local covariance approximation

In the diffusion benchmark, let R_0 be the existing fund return. Replacing a fraction δ of alternative a with investment j changes expected log growth approximately by

$$\Delta g = \delta\left[\mu_j - \mu_a - \operatorname{Cov}(r_j - r_a, R_0)\right] - \frac{\delta^2}{2}\operatorname{Var}(r_j - r_a) - \Delta c.$$

Consequently, the approximate hurdle is

$$\boxed{\mu_j - \mu_a \gtrsim \mathrm{Cov}(r_j - r_a, R_0) + \frac{\delta}{2}\mathrm{Var}(r_j - r_a) + \frac{\Delta c}{\delta}.}$$

This separates existing-portfolio exposure, the size of the new trade, and incremental costs.

It is not an exact pricing formula for a binary investment.

Why the existing private percentage matters for risk

Suppose an asset has unhedgeable annual volatility of 45%, independent of other positions.

If it already represents 45% of NAV, its own contribution to the marginal covariance charge is

$$0.45(0.45)^2 = 9.1\% \text{ per year.}$$

By contrast, starting from zero, buying a 5% position incurs an average own-variance charge of approximately

$$\frac{1}{2}(0.05)(0.45)^2 = 0.51\% \text{ per year.}$$

These are stylized diffusion calculations, but they show why **adding to a large existing position can require much more expected return than initiating a small position in the same asset**.

Hedgeable risk should be evaluated after feasible rehedging; otherwise you may charge for exposure that the public portfolio would remove.

The capital cost is marginal—not automatically the public book's average ROE

The alternative is right that financing constraints can be decisive. But "use the public book's return on equity" is too coarse.

Consider a public strategy with total-NAV exposure w:

$$g_\alpha(w) = \mu w - \frac{1}{2}\sigma^2 w^2.$$

Suppose its exposure is limited by liquid capital:

$$w \le m\frac{L}{N},$$

where m is the permitted exposure per dollar of liquid capital.

Moving δN from liquid assets into a private investment reduces that exposure ceiling by $m\delta$.

When the ceiling binds, the marginal lost log-growth opportunity per dollar moved is

$$\boxed{\lambda_{\text{capital}} = m(\mu - \sigma^2 w).}$$

It is the lost expected return **minus the benefit of shedding public risk**.

For example, assume

$$\mu = 6\%, \qquad \sigma = 20\%, \qquad m = 2.$$

The unconstrained optimum is

$$w^* = \frac{0.06}{0.20^2} = 1.5.$$

At 15% private, the financing ceiling is $2(0.85) = 1.7$, so it does not bind.

At 60% private, the ceiling is $2(0.40) = 0.8$, so it does bind. The marginal capital cost is

$$\lambda_{\text{capital}} = 2[0.06 - 0.20^2(0.8)] = 5.6\% \text{ per year.}$$

That differs from the strategy's 12% expected excess return per dollar of liquid equity at the maximum financing ratio.

Use the fund's actual borrowing and collateral terms. Assign no financing value to a private asset unless usable financing is genuinely available; do not assume a universal collateral rule.

What the convex liquidity penalty implies for hurdles

For the illustrative penalty, at 60% private,

$$c'(0.60) = \frac{2f_0}{0.30}.$$

With $f_0 = 2\%$,

$$c'(0.60) = 13.33\%.$$

Increasing the private allocation from 60% to 61% raises the annual penalty from

$$2.000\% \quad \text{to} \quad 2.136\%$$

of total NAV.

The incremental cost is approximately 13.6 basis points of fund NAV annually, or 13.6% of the newly allocated capital.

That explains the alternative's large marginal liquidity charges. **They are generated by the assumed cost function, not discovered from market data.**

For an investment lasting several years, calculate the incremental cost along its actual path:

$$\Delta C_{\text{liq}} = \mathbb{E}\int_0^T \left[c(\pi_t^{\text{new}}) - c(\pi_t^{\text{baseline}})\right] dt.$$

If A becomes liquid after 18 months, a five-year C investment may impose most of its *high-allocation* cost during those first 18 months. But the states where A remains locked must also be included.

Do not double-count: foregone public opportunities, financing constraints, and a reduced-form liquidity penalty may describe overlapping costs. The final decision should come from one consistently optimized model.

8. Treat A, B, and C differently—but do not rank them by stage alone

Profile	Principal economic questions	Principal liquidity questions
A: mature, near-IPO	Entry valuation, concentration, factor risk, company-specific repricing, security terms	When can the fund actually sell or finance its position, including delayed-exit states?
B: late-stage	Business execution, financing, dilution, valuation, dependence on favorable capital markets	Does the lockup extend precisely when the rest of the fund needs cash? What additional funding may be needed?
C: early-stage, option-like	Full outcome distribution, probability uncertainty, dependence on other holdings, position size	How much capital is committed over its life? Can success produce a large but still unmonetizable holding?

Future capital commitments are a distinct risk from simply being unable to sell the current holding; private-allocation research models these commitment and funding features explicitly. (onlinelibrary.wiley.com)

A is not low-risk simply because business success is likely

At a 45% fund weight, a 50% loss in A costs

$$45\% \times 50\% = 22.5\%$$

of total NAV.

A successful business can still be a poor investment at a sufficiently high price.

B can have particularly adverse liquidity timing—but is not always worst

B is concerning when weak business outcomes, financing needs, and delayed liquidity occur together.

That is an important risk pattern. It does not establish that B is always more dangerous per dollar than A or C. Price, position size, funding obligations, and joint outcomes decide the comparison.

C's limited downside is valuable, but "option-like" does not mean uncorrelated

A fully paid 5% position with no additional obligations can lose at most 5% of NAV.

But failure generates no liquid cash. It reduces the private percentage by destroying wealth, not by improving funding resources.

Success can also raise the private percentage. In Setting 1, if C grows from 5 to 40 while everything else is unchanged,

$$P = 10 + 40 = 50, \qquad N = 85 + 10 + 40 = 135,$$

so

$$\pi = 37.0\%.$$

The fund is richer but more illiquid. Whether secondaries are available must be investigated, not assumed.

9. Work through C's IRR calculation properly

Use the alternative's payoff assumptions:

$$M = \begin{cases} 8, & \text{with probability } 0.30, \\ 0.1, & \text{with probability } 0.70, \end{cases}$$

realized after four years.

The expected terminal multiple is

$$\mathbb{E}[M] = 0.30(8) + 0.70(0.1) = 2.47.$$

Annualizing that expected multiple gives

$$r_{\text{EV}} = 2.47^{1/4} - 1 = 25.4\%.$$

This is neither the expected scenario IRR nor a risk-adjusted IRR.

Value a whole 5% position

Assume the remaining portfolio earns a certain 10% annually. This is a simplifying benchmark, not a forecast.

Its four-year multiple is

$$B = 1.1^4 = 1.4641.$$

Putting 5% in C produces terminal fund wealth

$$G = 0.95B + 0.05M.$$

Thus,

$$G_{\text{failure}} = 1.395895, \qquad G_{\text{success}} = 1.790895.$$

Expected log wealth is

$$\mathbb{E}[\log G] = 0.70 \log(1.395895) + 0.30 \log(1.790895) = 0.408290.$$

The certainty-equivalent terminal fund wealth—the certain amount giving the same log utility—is

$$G_{\text{CE}} = \exp(0.408290) = 1.504243.$$

Define the equivalent certain payoff multiple on the private allocation by

$$0.95B + 0.05M_{\text{CE}} = G_{\text{CE}}.$$

Then

$$M_{\text{CE}} = \frac{1.504243 - 0.95(1.4641)}{0.05} = 2.266958.$$

Therefore,

$$\boxed{r_{\text{CE}} = 2.266958^{1/4} - 1 = 22.7\%.}$$

That is the certainty-equivalent return of the entire 5% position, conditional on the assumed rest of the portfolio and before liquidity costs.

Value another dollar added to that position

Once the 5% position exists, adding more puts additional money into states in which the fund is already doing well.

The marginal certainty-equivalent terminal multiple is

$$\boxed{M_{\text{marg}} = \frac{\mathbb{E}[M/G]}{\mathbb{E}[1/G]}.}$$

This is obtained by finding the certain payoff that makes the expected utility derivative equal:

$$\mathbb{E}\left[\frac{M - M_{\text{marg}}}{G}\right] = 0.$$

For the example,

$$M_{\text{marg}} = 2.078164,$$

and hence

$$r_{\text{marg}} = 2.078164^{1/4} - 1 = 20.1\%.$$

The existing position's average attractiveness and the attractiveness of adding more are different.

Diversification across five genuinely independent names

Split the same total 5% allocation equally across five independent investments with identical payoff distributions.

If $K \sim \text{Binomial}(5, 0.30)$, their average payoff multiple is

$$\overline{M} = 0.1 + 7.9\frac{K}{5}.$$

Apply the same formulas using

$$G = 0.95B + 0.05\overline{M}.$$

The results are:

Allocation	Annualized expected multiple	Whole-allocation certainty-equivalent IRR	Marginal IRR from increasing the same allocation pro rata
One name, 5%	25.4%	**22.7%**	**20.1%**
Five independent names, 1% each	25.4%	**24.8%**	**24.3%**

The diversification benefit is substantial—but depends on independence.

For your actual fund, the rest of the portfolio is risky, not a certain 10% investment. Replace G with simulated whole-fund wealth, including A, B, public positions, funding needs, and liquidity states.

Takeaway: there is no universal "C IRR haircut." Specify the rest of the portfolio, investment size, dependence, horizon, and whether the question concerns the whole position or an additional dollar.

10. What I would implement for this fund

I would use a joint portfolio process with three outputs.

An economic-risk view would include current-information private valuations, factor exposures, company-specific scenarios, concentration, and total-fund drawdowns. Reported return histories would not be the primary private-risk estimator.

A cash-capacity view would track usable resources across horizons and scenarios, including hedge losses, collateral changes, normal cash redemptions, commitments, and delayed exits. Build this as a reconciled cash-flow calculation so a derivative loss is not counted twice as both P&L and margin.

A marginal-investment view would compare the optimized fund with and without each proposed private investment, reporting the change in expected log wealth and the break-even entry price or payoff assumptions.

Two balance-sheet levers deserve explicit treatment.

With proportionate cash and private-asset distributions, redemptions can preserve the remaining fund's private percentage. Cash-only redemptions increase it. That is portfolio arithmetic; implementation depends on the fund documents and the treatment of investor claims.

Fundraising reduces the private fraction, but the amounts required can be large. Starting at 60% private, raising cash equal to f times existing NAV gives

$$\pi_{\text{new}} = \frac{0.60}{1+f}.$$

Reaching 50% requires new cash equal to 20% of existing NAV. Reaching 30% requires new cash equal to 100%. Given your stated experience, uncommitted fundraising should not support the adverse-state financing plan.

A secondary sale should likewise be evaluated as a whole-fund trade: compare the discount paid with the concentration reduction, liquidity benefit, changed public opportunities, and foregone private payoff. The alternative's annual hurdle estimates do not by themselves establish that a 5–10% discount is attractive.

Final takeaways

At 15% private: total NAV is the correct benchmark for sizing genuinely independent public alpha when prospective liquidity costs and financing constraints are small. This is not permission to ignore private economic exposures.

At 60% private: retain total NAV as the wealth objective, but derive an effective alpha capital base from the value of liquid wealth. The illustrative model supports a blend, but not a universal "public NAV plus half of private NAV" rule.

Do not give the entire public sleeve one volatility multiplier. Separate independent alpha from private-exposure hedges. A higher-volatility public sleeve can be the lower-risk choice for the fund.

For marginal privates: the hurdle depends on what the investment displaces, existing concentration, joint outcomes, financing capacity, and liquidity timing. A large position in A can be more consequential than a small binary position in C; B is not automatically the worst category.

Use IRR as a reporting summary, not the decision rule. The decision rule is the change in expected whole-fund log wealth after accounting for feasible financing and portfolio adjustments.

The compact operating principle is:

> **Count all private wealth in the objective, count all private economic risk in the portfolio, and credit private assets with liquidity only when—and to the extent—that liquidity is actually available.**